[Date]

The Editors, Applied Geography

Re: Submission of a research article, “Where could rice cultivation systems be transferred beyond traditional growing regions? A multi-scale gate analysis of thermal transfer, water access and food-supply risk on European arable land”

Dear Editors,

We submit a research article that asks a geographic question: where on existing arable land a cold-region rice cultivation system could be transferred, and how the answer changes with data source, seasonal window and spatial scale. Calibrating the Hokkaido thermal bound on the same grid where it is applied lowers it from ~900 to 589 GDD10. This moves the binding constraint for 106.9 Mha of European arable land from temperature to irrigation infrastructure and water allocation. Only 17.6% of that land combines thermal compatibility with existing irrigation infrastructure, and the classification of land near the bound depends on resolution.

The manuscript speaks to the journal's interest in spatially explicit resource assessment and land-use decision problems. It builds on land-suitability, cold-margin rice and irrigation studies published in Applied Geography and adds an explicit land denominator, non-compensatory gates and a scale-sensitivity analysis.

An earlier version was considered by Agricultural Systems and declined on scope grounds, with the advice to submit to a more land-use-oriented journal. This version has been substantially re-analysed. It corrects an error in the Lower Danube water-demand calculation (a 250,000 ha scenario requires ~241 m^3/s, 22% of Q99, not ~2%) and an implementation inconsistency in the portfolio-risk model. The candidate regions are framed as targets for next-stage feasibility assessment only.

The work is original, is not under consideration elsewhere, and has a single author. All code and derived outputs are openly available; the repository link is on the title page. The author declares no competing interests. Use of AI-assisted tools is declared in the manuscript. We intend to publish under the subscription model.

Suggested reviewers: [to be supplied by the author].

Sincerely,

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